



**NORTHERN UNIVERSITY OF
BUSINESS AND TECHNOLOGY
KHULNA**

Project Work plan

Course Name: Software Development I
Course Code: CSE 2216

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Project Title: Scientific Calculator

Course: Software Development LAB

Programming Language: JAVA

10-Week Work plan for Scientific Calculator Project

Week	Phase
1	Project Planning & Concept Finalization
2	Requirements Analysis & System Design
3	Environment Setup & Project Init
4	UI Layout Header & Display
5	Button Grid Implementation
6	Core Logic: Arithmetic & State
7	Scientific Functions Implementation
8	Special Features & Polish
9	Error Handling & Testing
10	Final Testing, Docs & Submission

Week 1: Project Planning & Concept Finalization

- Select project topic and objectives.
- Define calculator features and scope.
- Prepare initial project plan.

Week 2: Requirements Analysis & System Design

- Gather functional requirements.
- Design system workflow and UI layout.
- Create basic wireframe.

Week 3: Environment Setup & Project Initialization

- Install JDK and IDE.
- Configure Java Swing environment.
- Create project structure and main class.

Week 4: UI Layout Development

- Develop header section.
- Create display panel and status area.
- Apply theme and styling.

Week 5: Button Grid Implementation

- Add numeric, operator, and scientific buttons.
- Arrange buttons using GridLayout.
- Apply button styling.

Week 6: Core Logic Implementation

- Implement arithmetic operations.
- Handle input, operators, and results.
- Manage calculator state.

Week 7: Scientific Functions Implementation

- Add sin, cos, tan, log, ln functions.
- Implement square, square root, power, and percentage operations.
- Add π and e constants.

Week 8: Special Features & Enhancement

- Implement ANS, AC, CE, and Backspace.
- Add ON/OFF functionality.
- Improve user experience and display behavior.

Week 9: Error Handling & Testing

- Handle divide-by-zero and invalid inputs.
- Test scientific functions.
- Fix bugs and optimize performance.

Week 10: Final Testing & Documentation

- Perform complete system testing.
- Prepare project documentation and screenshots.
- Finalize and submit the project.